Cryptography Toolkit

USER MANUAL

Cryptography Toolkit is a Python-based application that demonstrates the implementation of **AES-256 encryption**, **RSA encryption/decryption**, and **SHA-256 hashing** through a graphical user interface.

System Requirements

BeforerunningtheCryptography Toolkit,ensureyour systemmeets the following requirements and that all dependencies are properly installed.

Operating System

Windows 10 / Windows 11

Python Version

Python 3.10 or higher

Required Library

PyCryptodome Library

Install all required dependencies by running the following command in your terminal:

```
pip install -r requirements.txt
```

Running the Application

Once dependencies are installed, launching the application is straightforward. Open a terminal window, navigate to the project directory, and execute the main module using Python.

Step 1 — Open Terminal

Navigate to the project directory using your terminal or command prompt.

Step 2 — Run the Application

Execute the following command to launch the graphical interface:

```
python -m src.main
```

Using the Application

The Cryptography Toolkit supports three cryptographic operations. Follow the steps below for each algorithm.

AES Encryption

1. Select **AES**.
2. Enter text in the Input box.
3. Enter a password.
4. Click **Encrypt**.
5. Encrypted text will appear in the Output box.

AES Decryption

1. Select **AES**.
2. Enter the encrypted text.
3. Enter the same password used during encryption.
4. Click **Decrypt**.
5. Original text will be displayed.

RSA Encryption

1. Select **RSA**.
2. Click **Generate RSA Keys**.
3. Enter text in the Input box.
4. Click **Encrypt**.
5. Ciphertext will appear in the Output box.

RSA Decryption

1. Select **RSA**.
2. Enter the ciphertext.
3. Click **Decrypt**.
4. Original text will be displayed.

SHA-256 Hashing

1. Select **SHA-256**.
2. Enter text in the Input box.
3. Click **Hash**.
4. The generated hash value will appear in the Output box.

Troubleshooting

If you encounter issues while using the Cryptography Toolkit, refer to the common problems and solutions below.

1

Application Does Not Start

The most common cause is missing dependencies. Install all required packages by running:

```
pip install -r
requirements.txt
```

2

RSA Operations Fail

RSA encryption and decryption require a key pair. Make sure to click **Generate RSA Keys** before performing any RSA operation.

3

Empty Output

Ensure that valid input data has been entered in the Input box before clicking Encrypt, Decrypt, or Hash.

Conclusion

The Cryptography Toolkit provides a simple platform for understanding encryption, decryption, hashing, and key generation concepts using **AES**, **RSA**, and **SHA-256** algorithms.

AES-256

Industry-standard symmetric encryption for securing data with a password-based key.

RSA

Asymmetric public-key cryptography enabling secure key exchange and digital signatures.

SHA-256

Cryptographic hash function producing a fixed 256-bit digest for data integrity verification.